

Jerome, Yat Hei Siu

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📅 Available to work immediately

🇬🇧 Right to work in the UK

Profile

Cybersecurity Master's graduate (MSc, University of Birmingham) and Computer Science graduate operating with the full technical capabilities of a Network Security & Systems Engineer. Leverages a rigorous foundation in decentralized systems and network exploitation with hands-on experience in reverse engineering in IoT architecture. Fully eligible to work in the UK without sponsorship (BNO Visa) and equipped to deliver technical value in a junior penetration testing, security analyst, role.

Key Skills

- Proficient in Python, Java, PowerShell
- Proficient in Linux and Windows command-line environments
- Penetration Testing & Vulnerability Assessment
- Security Incident Root-Cause Analysis & Threat Modelling
- Decentralized Systems Architecture & Blockchain Technology
- Familiar with Industry-Standard Security Tools *Wireshark, Nmap, Metasploit*
- RF Signal Analysis and Familiar with Software-Defined Radio (SDR)
- Advanced Microsoft Office Environment Capabilities

Education

09/2024 - 01/2026 Msc Cyber Security, University of Birmingham

09/2022 - 07/2024 Bsc (Hons), Computer Science, University of Sussex - First Class Honours

Recent Projects

Security Analysis of Automotive Remote Keyless Entry (RKE) Systems:

- Engineered a black-box security analysis of a modern Mini Cooper RKE system using Software-Defined Radio (SDR) hardware to capture and reverse-engineer RF communications.
- Analyzed and Demodulated FSK signals and decode Manchester II data streams.
- Executed a replay attack that bypassed a 48-bit rolling code system, uncovering a synchronization vulnerability and a denial-of-service (DoS) anti-replay lockout mechanism.

Network Exploitation & Device Reverse Engineering on Apple TV:

- Analyzed wireless network traffic in monitor mode using Wireshark to investigate WPA encryption, and attempted Man-in-the-Middle (MitM) attacks utilizing Bettercap.
- Performed root-level filesystem analysis by establishing an SSH connection into a jailbroken Apple TV 4, subsequently extracting and reverse-engineering a third-party jailbreak application.

Blockchain-Integrated Smart Home IoT Architecture:

- Architected a decentralized smart home network utilizing Ethereum smart contracts (Solidity) to ensure tamper-proof data integrity and security for connected IoT sensors.
- Developed a Python-based MQTT-SmartContract bridge and a lightweight Mosquitto broker to facilitate real-time, low-bandwidth communication between the blockchain network and edge devices.
- Deployed a locally simulated blockchain environment using Ganache and created a dynamic web dashboard utilizing Web3.js for secure user authentication and live device telemetry.

References:

Dr. Marius Muench, University of Birmingham

m.muench@bham.ac.uk

Prof. Martin White, University of Sussex

m.white@sussex.ac.uk

Additional Information

Mobility: Full UK Driving License. (Willing to commute/relocate)

Languages: English (Fluent) IELTS 7.0, Cantonese (Native), Mandarin (Fluent).